

Design Document for SpotMe

Overview

In this milestone, I will implement a multi-agent workflow to SpotMe to provide personalized fitness guidance based on user input. Instead of relying on a single response, the system breaks the problem into smaller tasks handled by specialized agents. This allows the system to generate more accurate and useful feedback for users. The workflow is orchestrated by a Coordinator Agent, which routes the user's request to the appropriate agents and ensures the final response is clear and actionable.

Agents & Their Roles

1. Coordinator Agent (Planner)

The Coordinator Agent interprets the user's input and determines the type of request. For example, learning an exercise vs. fixing form issues. It routes the request to the appropriate agents and manages the workflow.

Input:

User responses (exercise, goal, issue)

Output:

Structured task instructions for other agents

2. Exercise Explanation/Diagnostic Agent

This agent handles the first stage of analysis.

- If the user is learning an exercise, it provides step-by-step explanation
- If the user is not feeling the exercise correctly, it asks diagnostic questions and identifies issues

Input:

Exercise + user problem

Output:

Explanation or identified issue

3. Form Correction Agent

This agent provides recommendations to fix techniques.

Input:

Identified issue or exercise context

Output:

Specific adjustments like stance, posture, tempo, weight

4. Muscle Activation Agent

This agent explains which muscles should be engaged and how the movement should feel.

Input:

Exercise + corrected form

Output:

Expected muscle activation cues

Milestone II – Multi-agent Workflow

5. Safety Agent (Critic)

This agent acts like a verifier to make sure that the guidance is safe and also realistic.

Input:

All previous outputs

Output:

Warnings, limitations, or corrections

6. Summary Agent

This agent combines all outputs into a concise, user-friendly response.

Input:

Outputs from all agents

Output:

Final structured response shown to the user

Also, the system can be extended to use external fitness APIs or knowledge bases for exercise validation and accuracy.

Coordination Mechanism

The system uses a fixed pipeline workflow:

Coordinator → Explanation/Diagnostic → Form Correction → Muscle Activation → Safety → Summary

This makes sure that each step builds on the previous one in a logical order.

Communication Between Agents

Each agent passes structured information to the next agent. Here's an example:

- Coordinator sends task type and context
- Diagnostic agent sends identified issue
- Form correction agent sends adjustments
- Safety agent reviews all outputs before finalization

This creates a clear flow of information.

Handling Failures or Low-Quality Outputs

- The Safety Agent flags potentially unsafe or unclear recommendations
- The Coordinator Agent can simplify responses if the user input is too vague
- The system avoids making overly confident claims when information is limited
- Users can provide follow-up input if guidance does not resolve the issue

Updated Code

Because I used Opal, there is no traditional hand-written code file. The multi-agent workflow was implemented directly in the Opal platform.

Milestone II – Multi-agent Workflow

Risks	Mitigation Plan	
	Design & development of agent	Agent integration into processes
Adversarial attacks	Adversarial training and action verification	Multi-step authentication, auditing, and real time monitoring of outputs
Bias and discrimination	Diverse training data, debiasing algorithms, and bias detection	Decision auditing
Misinformation and hallucinations	Verification mechanisms and confidence thresholds	Safety Agent validation, human oversight, and fallback procedures
Data privacy violations	Data governance and sanitization, and privacy-preserving mechanisms	Real-time privacy monitoring and decision auditing systems
Intellectual property risks	Dataset auditing and content tracking systems	Automated license verification systems and decision trees